

Impact-Aware State Estimation for Off-Road Motion: EKF vs. EKF+LSTM (IMU-Only) with a VIO Extension

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I. OVERVIEW

Off-road motion induces shocks, slip, and rapid attitude changes that violate the smooth-noise assumptions of standard inertial estimation, increasing drift and failure rates. Traditional IMU-based filters such as the Extended Kalman Filter (EKF) assume approximately smooth dynamics and fixed process noise, which can lead to overconfidence and degraded performance in highly dynamic motion [1]. To address similar challenges in time-series estimation, recent research has combined EKF with learning components such as LSTMs to adapt noise covariance online, yielding improvements in robustness under non-stationary conditions [2].

We will use the ROOAD dataset [1], which provides monocular images, IMU measurements, and ground-truth trajectories collected in challenging off-road environments. High-dynamic segments will be identified using IMU jerk metrics, allowing focused evaluation of impact-prone motion. Our approach compares a baseline IMU-only EKF with an impact-aware EKF augmented by an LSTM that adaptively scales process noise. The primary scope is IMU-only; a stretch extension applies the same idea to IMU+camera visual-inertial odometry (VIO).

The objective of this project is to implement and compare (1) a baseline IMU-only EKF and (2) an impact-aware EKF augmented with an LSTM that adapts filter uncertainty online. As an extension (time permitting), we will apply the same idea to IMU + camera VIO, to test whether learned uncertainty adaptation improves stability in visually challenging, high-dynamic segments.

II. PROBLEM STATEMENT

A conventional IMU EKF assumes approximately smooth dynamics with fixed process noise, becoming overconfident under impulsive off-road motion. Our key question is: *Does an LSTM-driven adaptive process-noise EKF reduce drift and failures on ROOAD compared to a fixed-noise EKF (IMU-only), and can the same idea improve VIO stability (stretch)?*

III. TECHNICAL APPROACH

We will use the ROOAD dataset, which provides synchronized IMU data, monocular images, and ground-truth trajectories collected in off-road environments. Ground truth will be converted to a local metric frame for evaluation, and

high-dynamic segments will be identified using an IMU jerk metric to analyze impact-prone motion.

Our baseline will be an IMU-only Extended Kalman Filter (EKF) with state variables for orientation, velocity, position, and IMU biases. The filter will use fixed process noise and serve as a reference for performance under rough terrain conditions.

We propose augmenting the EKF with a lightweight LSTM that adaptively scales the process noise based on short windows of IMU data and jerk features. The network will predict a bounded noise scaling factor trained offline using ground-truth trajectory error. This preserves EKF interpretability while enabling impact-aware uncertainty adaptation.

As a stretch goal, we will extend the framework to include monocular visual updates and evaluate whether adaptive noise scaling improves visual-inertial robustness. Performance will be assessed using trajectory error metrics and robustness analysis on high-dynamic segments.

IV. EXPECTED OUTCOME

- IMU-only EKF baseline on ROOAD sequences.
- EKF+LSTM adaptive-noise estimator with improved robustness off-road.
- Quantitative comparison: EKF vs EKF+jerk-based noise inflation vs EKF+LSTM.
- (Stretch) Improved stability/accuracy for IMU+camera VIO using the same adaptive module.

V. MILESTONES AND TIMELINE

- **Week 1:** ROOAD extraction; verify timestamps/topics; IMU-only EKF on 1 sequence.
- **Week 2:** Heuristic impact baseline (jerk-triggered Q scaling); evaluation pipeline (ATE/RPE + segment analysis).
- **Week 3:** Training windows + best- α labels; train LSTM; integrate EKF+LSTM (Q_k).
- **Week 4:** Ablations across sequences; finalize IMU-only results; *stretch*: add VIO update if time allows.

VI. TEAM ROLES

- **Pranav Jain:** ROOAD preprocessing; IMU-only EKF; heuristic baseline; evaluation/plots.
- **Avantika Rattan:** LSTM features/labels; model training; EKF+LSTM integration; ablations/robustness analysis.

REFERENCES

- [1] Unmanned Lab, "ROOAD: Off-road robotics dataset," <https://github.com/unmannedlab/ROOAD>, 2025.
- [2] Hybrid EKF-LSTM framework with GA-tuned noise covariance for enhanced time series forecasting, *Engineering Research* (2025). DOI:10.1088/2631-8695/adf9be.